25900656

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025
THIRD SEMESTER

VAC - MG3VACSTA200 - APPLIED STATISTICAL ANALYSIS : ETHICAL DATA COLLECTION, INTERPRETATION AND DECISION MAKING IN SOCIETY

SCHEME OF VALUATION

PART-A

*Answer any **seven** Questions. Each question carries **two** marks.*

1. No linear relationship $\rightarrow$ 2 marks.
2. Definition of regression $\rightarrow$ 2 marks.
3. Definition of mode $\rightarrow$ 2 marks.
4. A graph, which represents the cumulative frequencies of a frequency table $\rightarrow$ 2 marks.
5. A statistical method used to determine if there is a significant association between two categorical variables. $\rightarrow$ 2 marks.
6. Give 2 marks for writing any test statistic, which is used for testing the equality of means or difference of means of two populations.
7. Definition of statistical inference $\rightarrow$ 2 marks.
8. $+1$ $\rightarrow$ 2 marks.
9. Any measure of central tendency $\rightarrow$ 2 marks.
10. Informed consent - Understanding about purpose, procedures, risks and benefits of the interview. Give 2 marks if atleast one of these terms is in the answer.

PART-B

*Answer any **four** questions. Each question carries **six** marks.*

11. Two attributes A and B are said to be associated if they are not independent. $\rightarrow$ 3 marks.

    If $\delta > 0$, the association is positive and if $\delta < 0$, the association is negative. $\rightarrow$ 3 marks.

12. The four measures are nominal, ordinal, interval and scale $\rightarrow$ 6 marks.

13. $\binom{n}{r} \times 2^r$ $\rightarrow$ 6 marks.

14. $H_0 : p = 0.5, \qquad H_1 : p \neq 0.5$ $\rightarrow$ 2 mark.

    $\hat{p} = \dfrac{55}{120} = 0.46$ $\rightarrow$ 1 mark.

    Test statistic $t = \dfrac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} \sim N(0,1)$ $\rightarrow$ 1 mark.

    $t = -0.913$ and $t_{\alpha/2} = 1.96$ $\rightarrow$ 1 mark.

    Conclusion : Failed to reject H_0 $\rightarrow$ 1 mark.

15. $H_0 : \mu = 50, \qquad H_1 : \mu \neq 50 \quad \rightarrow 2$ marks.

$t = \dfrac{(\overline{x} - \mu_0)\sqrt{n}}{\sigma} \sim N(0,1) \quad \rightarrow 2$ marks.

$t = 2$ and $t_{\alpha/2} = 1.96 \quad \rightarrow 1$ mark.

Conclusion: Reject the null hypothesis $\quad \rightarrow 1$ mark.

16. If the p-value is less than the significance level of the test, reject the null hypothesis and otherwise there is no evidence in the sample to reject the null hypothesis $\quad \rightarrow 6$ marks.

PART-C

*Answer any **one** questions.Each question carries **12** marks.*

17. Formula for finding the standard deviation or combined standard deviation of observations $\rightarrow 12$ marks.

18. $H_0 : \mu_1 = \mu_2, \qquad H_1 : \mu_1 \neq \mu_2 \quad \rightarrow 3$ marks.

$t = \dfrac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1) \quad \rightarrow 4$ marks.

$t = -13.4, \quad t_{\alpha/2} = 2.02 \quad \rightarrow 3$ marks.

Conclusion : Reject the null hypothesis. $\quad \rightarrow 2$ marks.

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